Microeconomics

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Artificial Intelligence and Financial Technologies
New Economic School · 2026/27

Economics is bigger than this course

One shared question, different lenses: **what shapes wages?**

CLASSICAL

Value and distribution
Land rent; wages and profits
Surplus and accumulation

Pay: shares of total output.

MARXIAN

Labour and surplus value
Exploitation; accumulation
Class and ownership

Pay: class and bargaining.

NEOCLASSICAL

Consumer and firm
Risk; intertemporal choice
Topkis: monotone statics

General equilibrium
Games and information

Pay: productivity and scarcity.

AUSTRIAN

Market process
Dispersed knowledge
Entrepreneurship

Pay: discovery and coordination.

KEYNESIAN

Effective demand
Post-Keynesian uncertainty
Financial instability

Pay: demand and employment.

INSTITUTIONAL

Rules and organisations
Property rights
Transaction costs

Pay: institutions and bargaining.

BEHAVIOURAL

Bounded rationality
Prospect theory
Social preferences

Pay: norms and reference points.

EVOLUTIONARY

Routines and selection
Innovation
Path dependence

Pay: changing technologies.

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→ An Amazon pricing case: mechanism, evidence, counterfactual

The course at a glance

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Final grade $= 0.20\,HW + 0.10\,Project + 0.70\,Exam$.

Each homework has two parts

1. **A mock exam.** Exam-level difficulty and length. Try it honestly in one **120-minute sitting**. The TA will mark it as a real exam so that you can assess your readiness. Errors in this part do not reduce your final course grade.

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Discuss approaches, but submit your own reasoning. If you use generative AI, disclose the tool and its role and be ready to explain every step.

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Topic after lecture 2; mechanism after lecture 4; final essay after lecture 6, before lecture 7. Presentations and discussion at seminar 7.

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The date, detailed format and permitted materials will be announced in advance.

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- Ask during class when a definition, assumption or argument is unclear.

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	Lecture	Seminar / milestone
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7	Intertemporal choice and finance	Saving, borrowing; project presentations; HW 3

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FTC: complaint and alleged anti-discounting practices

→ Back to why microeconomics

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Part II: useful assumptions

- Continuity and indifference curves
- Monotonicity, LNS and convexity
- Separability
- Quasilinearity

To model a choice, we need to know...

What do people **want**?

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What **can** they do?

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Start with what they want.

What is the primitive?

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- Quality, terms, timing and relevant context belong in the definition of an alternative.

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It is a statement about a pair of alternatives, not a numerical score or an observed purchase. Some introductory textbooks also introduce $\sim$ or $\succ$ separately. Do we need them as additional primitives?

Can we define indifference using only $\succeq$?

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Both comparisons hold. This differs from incomparability, where neither comparison holds.

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Does “not $x \succ y$” imply “$y \succ x$”?

No: $x \sim y$ is already a counterexample. Without completeness, incomparability is another.

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This does not require selfishness, perfect information or easy computation.

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Suppose the shared dishes, prices and information are identical, and each observed choice is the **unique favourite**. Can one fixed ranking explain both?

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For relation 2, try $(3, 2, 1)$, $(2, 1, 3)$ and $(1, 3, 2)$.

Condorcet’s paradox: rational voters, cyclic majority

Each voter has a complete and transitive ranking. Compare alternatives by simple majority.

Condorcet's paradox: rational voters, cyclic majority

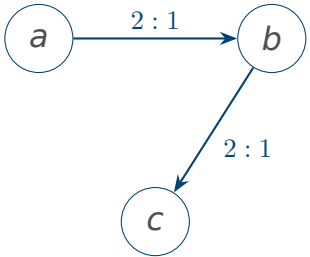
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3	$b \succ_3 c \succ_3 a$

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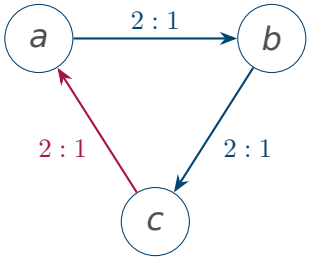
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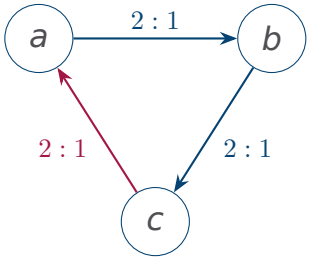
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Individual rationality need not produce a transitive collective ranking. No alternative beats every other alternative by majority.

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What must we observe besides a purchase before inferring anything about preferences?

Choice as the primitive

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A best element need not exist on an infinite menu. Nonemptiness is an additional requirement.

Consistency across menus: WARP

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NES

Weak Axiom of Revealed Preference

For any menus A, B and any $x, y \in A \cap B$,

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- One cannot be strictly beaten in one menu and selected over its rival in another.
- This formulation concerns the **whole correspondence**, including ties.

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Menu $A = \{a, b, c\}$

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a is selected; b is available but not selected.

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Preferences and choice: an equivalence

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Let X be finite and C be nonempty on **every** nonempty subset of X.
Then C satisfies WARP if and only if there exists a complete, transitive relation $\succeq$ such that

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$\rightarrow$ Proof, if time permits $\rightarrow$ Preferences $\rightarrow$ WARP

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If we only know that an observed choice is optimal, universal indifference can rationalise any such observations. Additional structure matters.

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$u : X \to \mathbb{R}$ represents $\succeq$ if, for every $x, y \in X$,

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- Representation turns choice into an optimisation problem.

Representation on a finite set

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The ranking matters. The gaps between the labels do not.

Ordinal, not a happiness meter

Alternative	u	v
A	3	100
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C	1	-20

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Both functions represent $A \succ B \succ C$.

Increasing transformations preserve the ranking

If f is strictly increasing on $u(X)$, then $v = f \circ u$ represents the same preferences:

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Examples: $u + 5$, $30u$, e^u. Squaring works on a nonnegative range, not in general. Why is a **weakly** increasing transformation not enough?

No interpersonal comparisons for free

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An individual ordinal representation does not, by itself, supply interpersonal units or a social welfare criterion.

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The non-representation proof uses uncountably many disjoint intervals; see the appendix.

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4. What changes when we move from finite to infinite choice sets?

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Lexicographic counterexample: $(1/n, 0) \succ_{\text{lex}} (0, 1)$ for all n, but $(0, 0) \prec_{\text{lex}} (0, 1)$ in the limit.

Indifference curves: equally good bundles

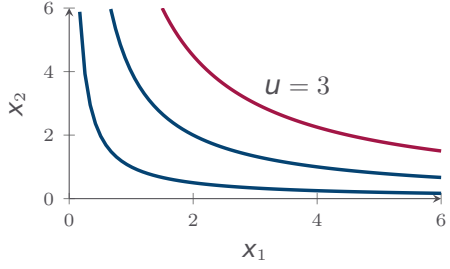
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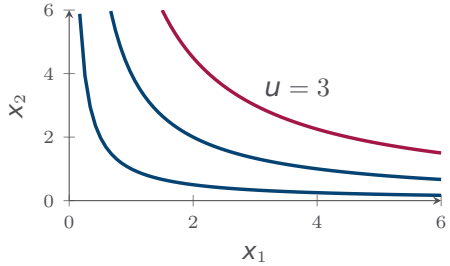
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These are level sets for the example $u(x) = \sqrt{x_1 x_2}$.
Different utility labels can describe the same indifference curves.

Monotonicity: fix the notation first

Write $x \geq y$ if $x_i \geq y_i$ for every i, and $x \gg y$ if $x_i > y_i$ for every i.

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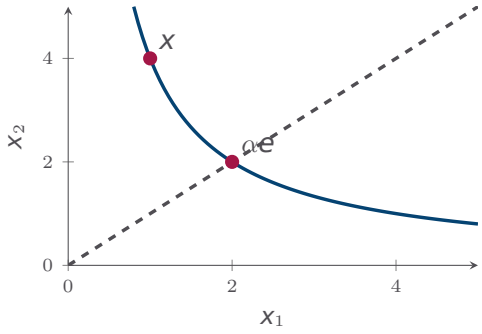
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- The assumption is inappropriate for a bad unless we redefine the commodity.

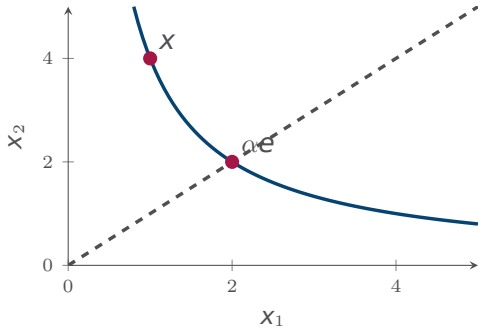
A constructive picture: the diagonal bundle

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Now add monotonicity to rationality and continuity. Each x is indifferent to a unique diagonal bundle αe, where $e = (1, \ldots, 1)$.

$$u(x) = \alpha \quad \text{if } x \sim \alpha e.$$

A useful way to **construct** a utility scale.

→ Proof under monotonicity

Local nonsatiation: a better bundle nearby

Coffee and cake: could there be a bundle where every sufficiently small change makes you no better off?

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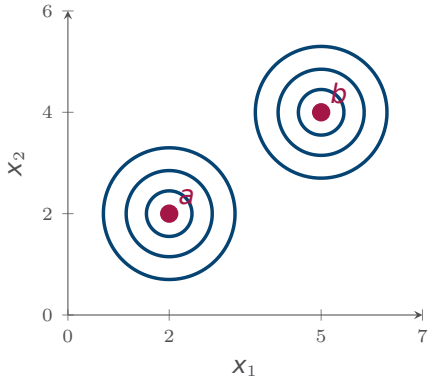
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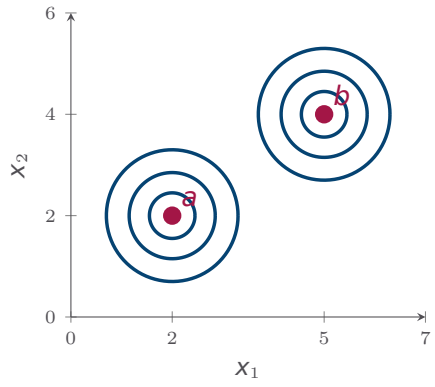
Example

$u(x_1, x_2) = x_1$ is locally nonsatiated but not strongly monotone.

Two bliss points



Two bliss points



Can these preferences be rational and continuous? Locally nonsatiated? Convex?

Convex preferences: mixing bundles

Assume X is convex. Preferences are **convex** if, for every $\lambda \in [0,1]$,

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This is a mixture of quantities, not a lottery over the two bundles.

Convex functions and convex preferences

On a convex domain, u is a **convex function** if

$$u(\lambda x + (1-\lambda)y) \leq \lambda u(x) + (1-\lambda)u(y), \qquad \lambda \in [0,1].$$

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These two examples do not establish the general claim. Which property of utility actually captures convex preferences?

Convex preferences and quasiconcavity

A convex representation need not exist. For $u(x) = \sqrt{x_1 x_2}$,

$$a = (1,4) \sim b = (4,1), \qquad m = \tfrac{1}{2}(a+b) = (2.5, 2.5) \succ a.$$

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contradicting $m \succ a$.

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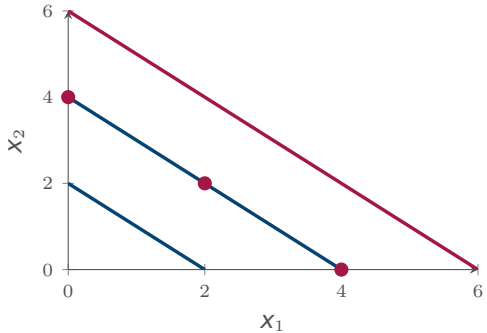
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Convex preferences $\iff$ quasiconcavity of a representing utility. This property survives strictly increasing transformations.

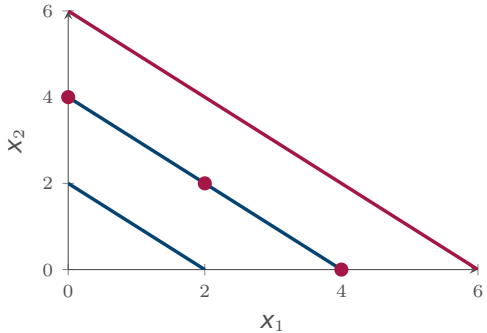
Perfect substitutes: a straight-line ranking

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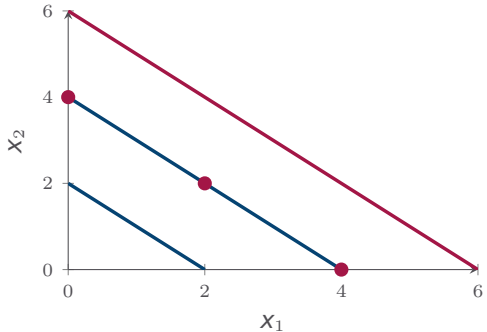
Perfect substitutes: a straight-line ranking



$$u(x) = x_1 + x_2.$$

One unit of either good contributes the same amount to the ranking.

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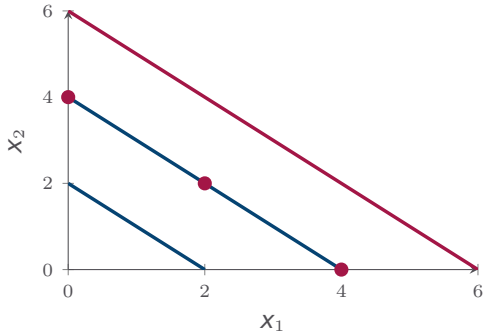


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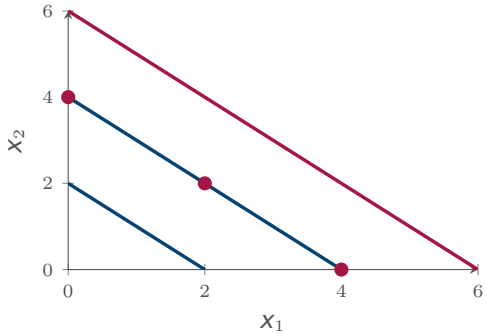


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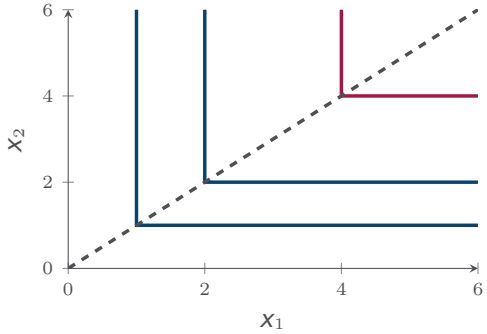
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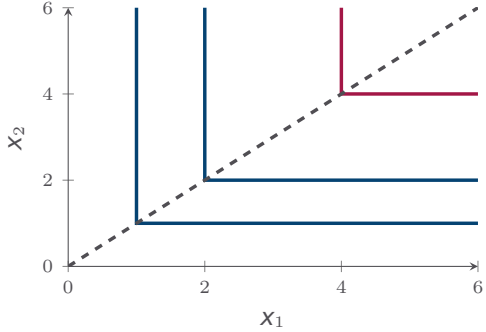
Perfect complements: useful in fixed proportions

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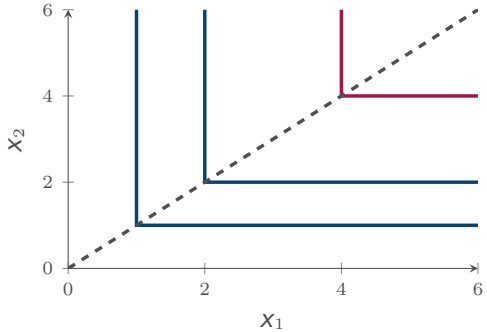
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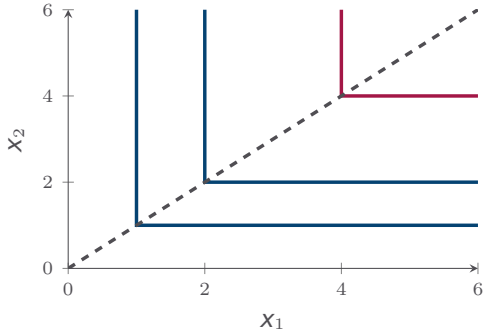


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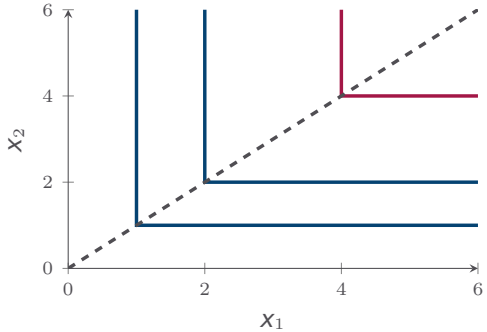


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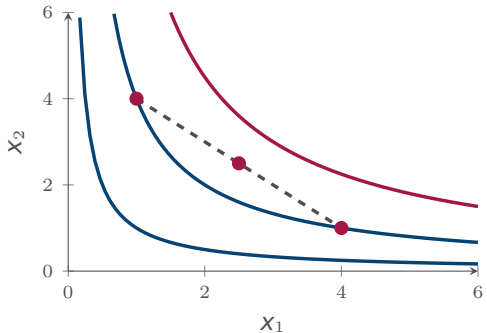
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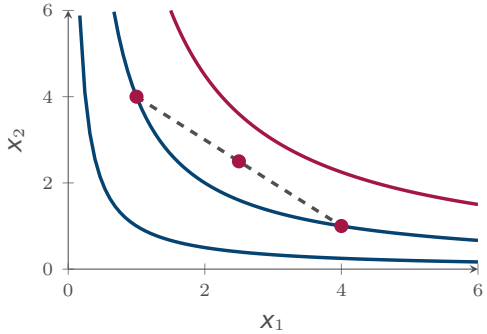
Cobb–Douglas: mixing can help

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$$u(x) = \sqrt{x_1 x_2}.$$

$u(1,4) = u(4,1) = 2$, but

$$u(2.5, 2.5) = 2.5.$$

On $\mathbb{R}^2_{++}$: strongly monotone and strictly convex preferences.
On $\mathbb{R}^2_{+}$, axis bundles have zero utility; the strict claims fail at the boundary.

Separability: a ranking inside a larger bundle

Let $X = Y \times Z$. Think of Y as food and Z as housing and transport.

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- The condition is not symmetric between Y and Z.

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At **any fixed** z, the first food bundle ranks higher. This does not fix the food budget. → Formal construction

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A convenient assumption with economic content: money compensation is independent of the starting numeraire level.

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Let $u(a, y) = a + 4\sqrt{y}$, with both prices equal to 1 and $a + y \leq m$.
Substitute the exhausted budget:

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12	4	8
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Thus $y^* = \min\{m, 4\}$. Demand for y is income-independent when the numeraire constraint is slack; income matters when that constraint binds.

What can they do? The budget set

For $p_1, p_2 > 0$ and income $m > 0$,

$$B(p, m) = \{x \in \mathbb{R}^2_+ : p_1 x_1 + p_2 x_2 \leq m\}.$$

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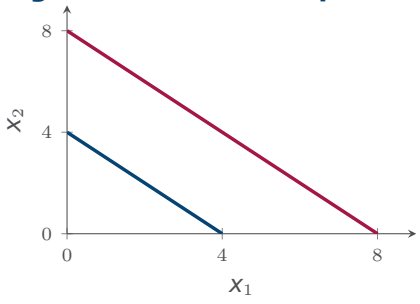
A higher p_1 at fixed p_2, m: a rotation inwards around the vertical intercept.

Preferences describe wants; the budget describes possibilities. Next lecture: solve the consumer's optimisation problem.

Budget geometry: income versus a price change

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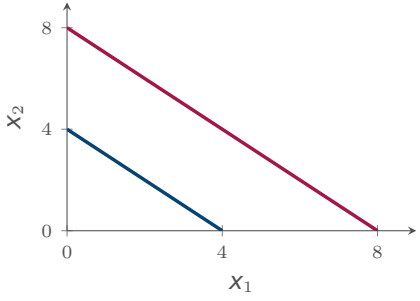
Higher income, fixed prices



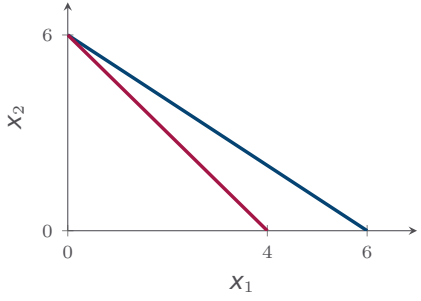
Budget geometry: income versus a price change

NES

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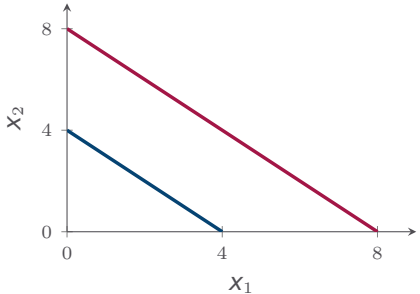


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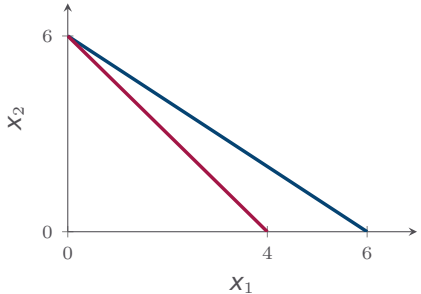


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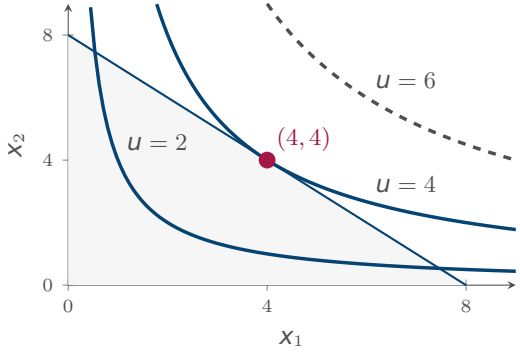
Higher p_1, fixed p_2, m



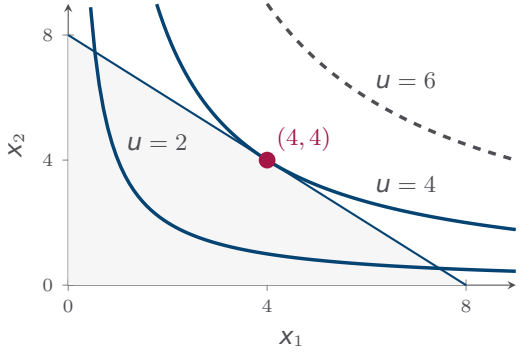
The feasible set expands in the first case and shrinks in the second. Neither change is itself a change in preferences.

Wants and possibilities in one picture

Wants and possibilities in one picture



Wants and possibilities in one picture



Possibilities:

$p_1 = p_2 = 1$, $m = 8$.

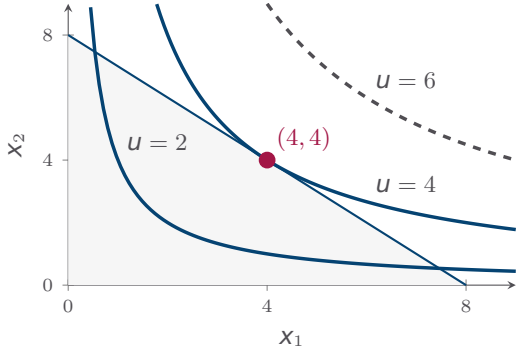
$x_1 + x_2 \leq 8$.

Wants:

$u(x) = \sqrt{x_1 x_2}$.

The best affordable bundle is $(4, 4)$, with utility 4.

Wants and possibilities in one picture



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The best affordable bundle is $(4, 4)$, with utility 4.

Why is $u = 6$ unattainable? Why settle for $u = 2$?

Exit ticket

1. Why can a strictly increasing transformation change concavity but not preferences?

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5. What is the boundary caveat for quasilinear demand?

Use a definition, example or short argument for each answer.

What to keep, and what to read

NES

Main reading

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Chapter 1: *Preference and Choice*; Chapter 3: *Classical Demand Theory*, sections 3.B–3.C on preferences and utility.

Appendix: why rational preferences imply WARP

Suppose $x, y \in A \cap B$, $x \in C(A)$ and $y \in C(B)$.

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Appendix: why rational preferences imply WARP

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Suppose $x, y \in A \cap B$, $x \in C(A)$ and $y \in C(B)$.

1. Because $y \in A$, optimality of x gives $x \succeq y$.

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3. For every $z \in B$, $y \succeq z$; transitivity gives $x \succeq z$.

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The argument uses transitivity and the full best-alternative interpretation of C. Finite menus ensure nonemptiness separately.

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Appendix: reconstructing the relation from choice

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- **Transitivity:** let $x \succeq y \succeq z$ and choose $w \in C(\{x, y, z\})$. WARP with the relevant pair menus first brings y or x into the triple, and then brings x into it. WARP with $\{x, z\}$ gives $x \succeq z$.

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Domain: all nonempty subsets of a finite X. Repeated alternatives in a triple give trivial cases.

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Appendix: limited data can hide a cycle

Suppose only pair menus are observed:

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$$C(\{s,p\}) = \{s\}, \qquad C(\{p,r\}) = \{p\}, \qquad C(\{s,r\}) = \{r\}.$$

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Passing WARP on an arbitrary observed domain does not imply rationalisability on the full domain.

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Appendix: why lexicographic utility does not exist

Suppose a real-valued u represents lexicographic preferences on $[0,1]^2$.

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2. Thus $I_t = (u(t,0), u(t,1))$ is a nonempty open interval.

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There cannot be an injection from uncountably many $t \in [0,1]$ into the countable rationals. Contradiction.

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Appendix: construction under monotonicity

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2. On $[0, M]$, $M = \max_i x_i$, the sets $\{a : ae \succeq x\}$ and $\{a : x \succeq ae\}$ are nonempty, closed and cover the interval. Connectedness makes them intersect.

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4. Transitivity and diagonal monotonicity give $x \succeq y \iff u(x) \geq u(y)$.
5. If $x^n \to x$, the coefficients $u(x^n)$ are bounded. Every convergent subsequence has limit $u(x)$ by continuity and uniqueness. Therefore $u(x^n) \to u(x)$.

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Appendix: separability in one construction

Fix z_0 and let $v(y) = u(y, z_0)$. For $t \in v(Y)$, pick $\hat{y}(t)$ with $v(\hat{y}(t)) = t$, and define

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- If $t > t'$, strict preference at z_0 carries to every z, so $U(t, z) > U(t', z)$.

The constructed aggregator has domain $v(Y) \times Z$.

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Appendix: primitives behind quasilinearity

On $\mathbb{R}_+ \times Y$, assume rationality, a worst $\bar{y}$, and:

→ Back to the lecture

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$$(a, y) \succeq (a', y') \iff a + v(y) \geq a' + v(y').$$

So $u(a, y) = a + v(y)$ represents preferences.

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Appendix: LNS and budget exhaustion

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If $p \cdot x^* < m$, let $\delta = m - p \cdot x^* > 0$.

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$$p \cdot y \leq p \cdot x^* + \|p\| \|y - x^*\| < m.$$

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LNS gives a strictly better y within this neighbourhood. It is feasible, contradicting optimality.

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LNS implies budget exhaustion at an existing optimum; it does not establish existence.

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Appendix: increasing transformations and risk

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$$\mathbb{E}[e^A] = e^2 \approx 7.39 < \frac{1 + e^3}{2} \approx 10.54 = \mathbb{E}[e^B].$$

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Appendix: increasing transformations and risk

Compare $A = 2$ for sure with $B = 0$ or 3, each with probability $1/2$.

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After the strictly increasing transformation $f(t) = e^t$,

$$\mathbb{E}[e^A] = e^2 \approx 7.39 < \frac{1 + e^3}{2} \approx 10.54 = \mathbb{E}[e^B].$$

The ranking of individual outcomes is unchanged, but the ranking of lotteries by expected transformed numbers changes. Ordinal deterministic utility and expected utility require different care.

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Appendix: answers to the relation exercise

1. Sum of coordinates: complete and transitive, by the ordering of real numbers.

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Appendix: answers to the relation exercise

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2. At least two of three coordinates: complete, but not transitive. Let $a = (3, 2, 1)$, $b = (2, 1, 3)$, $c = (1, 3, 2)$. Then $a \succ b$, $b \succ c$, $c \succ a$.

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Appendix: answers to the relation exercise

NES

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2. At least two of three coordinates: complete, but not transitive. Let $a = (3, 2, 1)$, $b = (2, 1, 3)$, $c = (1, 3, 2)$. Then $a \succ b$, $b \succ c$, $c \succ a$.
3. Componentwise comparison: transitive, but incomplete. $(2, 1)$ and $(1, 2)$ are incomparable.

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